

Project Requirements

Students per Team: Min 4 - Max 6	Deadline: 2 May 2026
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1. Project Overview:

- The objective of this project is to design and implement an end-to-end **Automated Machine Learning Application**. The goal is to build an intuitive platform that allows non-technical users to upload raw data, select an ML task, and receive a fully trained and evaluated model, all without writing a single line of code.
- Your application must successfully implement a very simple Frontend (UI) and a Backend, fulfilling the following technical requirements.

2. Technical requirements:

A. Data Ingestion (Frontend):

- File Upload: Allow the user to upload a dataset. The application must support at least .csv and .xlsx (Excel) file formats.
- Data Preview (Optional but recommended): Display the first few rows of the uploaded dataset so the user can verify their upload.

B. Task Selection (Frontend):

- ML Problem Type: Provide a dropdown or radio button menu allowing the user to select the type of machine learning problem they want to solve:
 - i. Classification
 - ii. Regression
 - iii. Clustering
- Target Variable Selection (For Supervised Learning): If Classification or Regression is selected, prompt the user to select the target column (label) from the uploaded dataset.

C. Automated Data Pipeline & Modeling (Backend):

Once the user initiates the process, your backend must automatically execute the following sequence:

- Automated Data Preprocessing Pipeline (select one technique to implement for each preprocessing step):
 - i. Handle missing values.
 - ii. Encode categorical variables.
 - iii. Scale/Normalize numerical features.
 - iv. If the target column is imbalanced, perform any resampling technique.

D. Model Training:

- Automatically split the data into training and testing sets (e.g., 80/20 split).
- Implement at least two different algorithms per ML task type. Train both of them and select the best one.

E. Model Evaluation:

- Calculate appropriate metrics based on the task:
 - i. *Classification*: Accuracy, Precision, Recall, F1-Score, Confusion Matrix.
 - ii. *Regression*: Mean Absolute Error (MAE), Mean Squared Error (MSE), R^2 Score.

iii. *Clustering*: Silhouette Score.

F. Results & Export (Frontend)

- Display a clear, readable report of the model's performance metrics to the user. (Visualizations like feature importance or confusion matrices are highly encouraged).
- Add a "Save Model" button. When clicked, this should serialize the trained model (and the preprocessing pipeline) using pickle or joblib and allow the user to download it as a .pkl or .joblib file.

G. API Implementation:

- Use **FastAPI** or **Flask** to create endpoints for data upload, training, and model retrieval.
- Requirement: The frontend must communicate with these endpoints via HTTP requests.

Important notes:

- This is a group project of **4 to 6** members.
- **The team with less than 4 members will not be discussed**
- All team members should work and fully understand everything in the project even if you distributed the work, you should understand your colleagues' work.
- The deadline for project submission of the project source code on classroom is on **Saturday, 2nd of May 2026 at 11:95 PM**. No late submission is allowed.
- Add the source code to a folder and compress it to a .zip file. Rename the .zip file "firstStudID_SecondStudID_ThirdStudID_FourthStudID_FifthStudentID_SixthStudentID.zip". The compressed file would be the file to be delivered.
- **Any cheating will be graded ZERO for both teams.**
- **Any team member who misses attending the discussion will take ZERO.**